EDA (Phasic Skin Conductance) Signal

VRET pipeline • Writer: Shayan Itami

1. What this signal is, and why we use it

Electrodermal activity (EDA) — also called galvanic skin response (GSR) or skin conductance — measures the electrical conductance of the skin, which varies with eccrine sweat-gland activity. Sweat glands are innervated exclusively by the sympathetic nervous system, so EDA provides a direct, peripheral index of sympathetic arousal: emotional or cognitive stress reliably raises skin conductance.

A raw EDA signal has two components. The **tonic** component (SCL — Skin Conductance Level) is a slow, drifting baseline driven by hydration, temperature, and long-timescale arousal. The **phasic** component (SCR — Skin Conductance Response) is the fast, event-driven bump superimposed on that baseline; it reflects discrete sympathetic activations and is the signal that tracks moment-to-moment arousal. This pipeline discards the tonic level and uses only the phasic component, because the tonic drift carries almost no useful arousal information on the timescale of a VR exposure session.

EDA carries weight **0.5** in the composite stress score S_t (EDA 0.5, HRV 0.3, HR 0.2) — the largest weight — because EDA is the most direct peripheral marker of sympathetic activation relevant to phobia exposure.

■ ***Reference.*** *Boucsein, W. (2012). Electrodermal Activity (2nd ed.). Springer. — The reference text for EDA physiology, measurement, and the tonic/phasic distinction. Dawson, M. E., Schell, A. M., & Filion, D. L. (2017). The electrodermal system. In J. T. Cacioppo et al. (Eds.), Handbook of Psychophysiology (4th ed.). Cambridge. — Sympathetic innervation of sweat glands and EDA as an arousal index.*

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2. The signal path at a glance

Raw EDA (1000 Hz) → **raw buffer** (rolling 60 s deque) → **downsample to 10 Hz** (nk.signal_resample) → **clean** (nk.eda_clean) → **phasic decomposition** (nk.eda_phasic) → **last phasic value** (current SCR) → **plausibility clamp** (|val| ≤ 1.0 μS) → **delta_eda** → z-score → S_t.

The whole path runs through a single helper, **compute_phasic_eda()**, called identically during the baseline sweep and at every live compute tick.

■ ***Reference.*** *Makowski, D., Pham, T., Lau, Z. J., et al. (2021). NeuroKit2: A Python toolbox for neurophysiological signal processing. Behavior Research Methods 53:1689-1696. — The library providing eda_clean, signal_resample, and eda_phasic.*

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3. Configuration constants

The values governing the EDA computation, quoted from the source:

```
EDA_DECOMP_RATE_HZ     = 10    # phasic decomposition runs at 10 Hz
EDA_DECOMP_WINDOW_S    = 60    # rolling raw-EDA window fed to nk.eda_phasic
EDA_DECOMP_WIN_SAMPLES = EDA_DECOMP_WINDOW_S * SAMPLING_RATE_HZ
EDA_PHASIC_SIGMA_FLOOR = 0.02  # phasic-EDA baseline spread floor in uS
EDA_PHASIC_MAX_US      = 1.0   # plausibility ceiling for a single phasic value
SIGMA_THRESHOLD        = 3     # 3-sigma cleaning for the tonic baseline mean
```

EDA_DECOMP_RATE_HZ = 10. Raw EDA is sampled at 1000 Hz, but phasic skin-conductance responses are slow events — their content lives well below 1 Hz. Downsampling to 10 Hz before decomposition reduces the series to 600 points per 60 s window instead of 60,000, making the call to nk.eda_phasic roughly 100x cheaper with no loss of information. 10 Hz is sufficient because it is still ten times the Nyquist rate for the fastest real SCR.

EDA_DECOMP_WINDOW_S = 60. The phasic decomposition (a highpass filter) needs a surrounding context to separate the slow tonic drift from the fast phasic bumps. A 60 s window provides enough tonic context for a stable separation while remaining short enough to track changes during an exposure session.

EDA_PHASIC_SIGMA_FLOOR = 0.02 μS. The z-score divides by the phasic EDA baseline spread. If a flat baseline yields a near-zero spread, a tiny live phasic change becomes an enormous z-score. The floor refuses to believe the resting phasic spread is below 0.02 µS; it only engages on degenerate baselines and prints a [SIGMA-FLOOR] warning.

EDA_PHASIC_MAX_US = 1.0. A physiological ceiling on any single phasic value returned by the decomposition. Genuine phasic SCRs are in the tenths-of-a-microsiemen range; values above 1 µS are a known decomposition artifact (resampler edge ringing), not real arousal. This constant is the threshold of the plausibility clamp explained in §4.

SIGMA_THRESHOLD = 3. Used for 3-sigma cleaning of the raw EDA baseline mean (the tonic reference avg_eda). Samples more than 3 standard deviations from the mean are removed before the average is computed, so a brief movement artifact cannot skew the tonic reference.

> ■ ***Reference.*** *Boucsein (2012) for EDA bandwidth and the rationale for 10 Hz downsampling. The 60 s decomposition window is an engineering choice of this pipeline consistent with short-term EDA analysis practice. The 1.0 uS phasic ceiling is grounded in the typical SCR amplitude range reported in Boucsein (2012) and Dawson et al. (2017).*

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4. The shared EDA routine — compute_phasic_eda()

Both the baseline sweep and the live phase obtain phasic EDA through this single helper. It takes the raw rolling EDA buffer and returns the current phasic SCR value in µS, or 0.0 if the window is too short, the decomposition fails, or the result is implausible.

```
def compute_phasic_eda(raw_eda_window, src_rate=SAMPLING_RATE_HZ):
    a = np.asarray(raw_eda_window, dtype=float)
    if a.size < src_rate * 5:
        return 0.0
    ds = nk.signal_resample(a, sampling_rate=src_rate,
                            desired_sampling_rate=EDA_DECOMP_RATE_HZ)
    ds = nk.eda_clean(ds, sampling_rate=EDA_DECOMP_RATE_HZ)
    phasic = nk.eda_phasic(ds, sampling_rate=EDA_DECOMP_RATE_HZ)["EDA_Phasic"].values
    if phasic.size == 0:
        return 0.0
    val = float(phasic[-1])
    if not np.isfinite(val) or abs(val) > EDA_PHASIC_MAX_US:
        print(f"[EDA-CLAMP] implausible phasic {val:+.2f} uS rejected")
        return 0.0
    return val
```

Lines 2-3 — minimum window check. At least 5 seconds of raw EDA must be available before decomposition is attempted; fewer samples give the highpass filter too little context for a stable tonic/phasic separation. Returns the safe neutral value 0.0 (no phasic deviation) until the buffer fills.

Line 5-6 — nk.signal_resample. Downsamples the 1000 Hz raw EDA window to 10 Hz. This is where a known edge artifact exists: the resampler intermittently produces a spurious ~0 as its very last output sample, creating a fake cliff. NeuroKit's zero-phase highpass filter then rings backward across the tail in response to this cliff, and phasic[-1] can read the ringing artifact (up to 4-15 µS) rather than real physiology. The plausibility clamp at lines 11-13 contains the blast radius of this artifact.

Line 7 — nk.eda_clean. Applies a lowpass filter to the 10 Hz EDA signal, removing high-frequency noise before decomposition.

Line 8 — nk.eda_phasic. Decomposes the cleaned EDA into its tonic (SCL) and phasic (SCR) components using a highpass filter method. The tonic component — the slow electrode-hydration drift that carries almost no useful arousal signal on session timescales — is discarded. Only the phasic series is retained.

Line 11 — phasic[-1]. Takes the *last* value of the phasic series — the most recent point in time, representing the current phasic state at the end of the rolling window. This is also the point most exposed to the resampler edge artifact described above, which is why the clamp on the next line is essential.

Lines 12-14 — plausibility clamp ([EDA-CLAMP]). Any phasic value whose magnitude exceeds 1.0 µS is rejected and replaced with 0.0. Real SCRs are tenths of a µS; a value above 1.0 is the resampler/filter artifact, not physiology. The clamp is not a repair — the artifact is left in the decomposition — it simply stops a known-garbage value from reaching the stress score or the Unity balloon. A [EDA-CLAMP] log line is printed whenever this fires (~0.5% of ticks on tested recordings).

Why no EDA smoothing? A previous version of the pipeline applied an exponential moving average (EMA, alpha=0.005) to the raw EDA for display. This was removed because: (a) the smoothed value was visually indistinguishable from raw — EDA is already an inherently slow signal; (b) the EMA was used only for display and never fed the score, so it contributed nothing to the pipeline's output; and (c) after the phasic redesign, the real EDA processing happens entirely inside compute_phasic_eda(), not on the smoothed trace. Raw EDA is now shown directly.

■ ***Reference.*** *Boucsein (2012) for the tonic/phasic distinction and typical SCR amplitude ranges (tenths of uS). Makowski et al. (2021) for nk.eda_clean and nk.eda_phasic. The resampler edge artifact and the 1.0 uS ceiling are documented from empirical testing on validation recordings (Paria calm recording, VAR arousal recording) conducted as part of this pipeline's development.*

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5. Baseline computation (the frozen reference)

During the 120 s baseline phase the pipeline computes two EDA references: a tonic mean (avg_eda, used only for display) and the phasic baseline distribution (eda_mean, eda_sigma) used for z-scoring during the live phase.

Step A — tonic baseline mean (avg_eda).

```
eda_raw_for_mean = np.asarray(eda_buffer, dtype=float)
mu = eda_raw_for_mean.mean()
sd = eda_raw_for_mean.std()
eda_cleaned = eda_raw_for_mean[np.abs(eda_raw_for_mean - mu) <= SIGMA_THRESHOLD * sd]
avg_eda = eda_cleaned.mean()
```

The raw 1000 Hz EDA samples are 3-sigma cleaned (any sample more than 3 standard deviations from the mean is removed) before the mean is taken. This prevents a brief movement artifact or cable-touch from biasing the tonic reference. avg_eda is the subject's resting skin conductance level; it appears in the freeze-block printout and is available to the operator.

Step B — phasic baseline decomposition.

```
_eda_ds = nk.eda_clean(
    nk.signal_resample(eda_raw_arr, sampling_rate=SAMPLING_RATE_HZ,
                       desired_sampling_rate=EDA_DECOMP_RATE_HZ),
    sampling_rate=EDA_DECOMP_RATE_HZ)
eda_phasic_baseline = nk.eda_phasic(
    _eda_ds, sampling_rate=EDA_DECOMP_RATE_HZ)["EDA_Phasic"].values
```

The entire baseline EDA buffer is decomposed once into its phasic series. This is the same resample → clean → phasic pipeline as the live helper, applied to the full 120 s baseline rather than to a rolling window. The resulting phasic series is stored as eda_phasic_baseline and indexed by the sweep loop to collect per-window phasic values.

Step C — baseline sweep: collecting eda_mean and eda_sigma.

```
# inside the 60s-step baseline sweep:
_raw_eda_d.append(_phasic_at_sample(end))    # phasic uS at this window
...
eda_mean  = float(_raw_eda_d.mean())
eda_sigma = float(_raw_eda_d.std())
eda_sigma = max(eda_sigma, EDA_PHASIC_SIGMA_FLOOR, SIGMA_FLOOR)  # floor 0.02 uS
```

_phasic_at_sample(end). A small helper that looks up the pre-computed eda_phasic_baseline at the sample index corresponding to the current sweep window's end point. This gives a consistent index mapping between the 1000 Hz ECG sweep position and the 10 Hz phasic baseline array.

eda_mean and eda_sigma. The mean and standard deviation of phasic EDA across all baseline sweep windows. Because phasic EDA is tonic-free and centered near zero, eda_mean is typically close to 0 µS. eda_sigma captures the natural resting variability of phasic activity — how much the subject's phasic EDA fluctuates when at rest. Both are frozen at baseline end and used to z-score every live phasic value.

Sigma floor (EDA_PHASIC_SIGMA_FLOOR = 0.02 µS). If the baseline phasic spread is implausibly small, dividing by it would amplify even a tiny live phasic value into a huge z-score. The floor refuses to believe resting phasic spread is below 0.02 µS and prints a [SIGMA-FLOOR] warning if it engages.

■ ***Reference.*** *Boucsein (2012) for the tonic/phasic distinction and resting SCR variability. The 3-sigma cleaning convention is standard in signal-quality preprocessing; the sigma floor follows the same guardrail pattern applied to HR and HRV (documented in those signal PDFs).*

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6. Live computation

During the live phase, raw EDA samples are appended to a rolling deque on every tick. The phasic decomposition is called every HR_COMPUTE_INTERVAL ticks (~0.5 s):

```
# per-tick: accumulate into rolling buffer
eda_raw_live.append(sample[eda_ch])

# every 25 ticks (~0.5 s): recompute
phasic_eda = compute_phasic_eda(eda_raw_live, src_rate=SAMPLING_RATE_HZ)
...
delta_eda = phasic_eda    # uS (phasic, tonic-free)
```

eda_raw_live (deque, maxlen = EDA_DECOMP_WIN_SAMPLES). A rolling buffer capped at 60 s × 1000 Hz = 60,000 samples. As new samples arrive the oldest drop off the left; the buffer always holds the most recent 60 s of raw EDA. It is seeded fresh at live-phase start — no baseline samples carry over — so the decomposition always runs on recent data.

compute_phasic_eda(eda_raw_live). Called as described in §4. Returns the current phasic EDA in µS, or 0.0 if the buffer is too short or the value is implausible. For the first ~5 s (before the minimum window fills) phasic_eda is 0.0 — the safe neutral value — and contributes nothing to the score.

delta_eda = phasic_eda. Unlike HR and HRV, the EDA delta is not expressed as a percentage deviation from baseline. It is the raw phasic value in µS. This works because the phasic component is already tonic-free and centered near zero: eda_mean (the baseline phasic mean, typically ~0 µS) is subtracted in the z-score step, so the z-score correctly measures how large the current SCR is relative to resting phasic variability, in the same standard-deviation units as the other signals.

■ ***Reference.*** *Dawson et al. (2017) for the event-driven nature of SCRs and the practice of measuring phasic amplitude directly. Makowski et al. (2021) for nk.eda_phasic.*

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7. How EDA enters the composite score

This section is common to all three signal documents so each stands alone. Each signal is converted to a z-score against its own frozen baseline mean and spread, then weighted:

```
def zscore_s_t(eda_phasic_v, hr_pct, hrv_pct):
    zE = (eda_phasic_v - eda_mean) / eda_sigma
    zH = (hr_pct       - hr_mean)  / hr_sigma
    zV = (hrv_pct      - hrv_mean) / hrv_sigma
    return WEIGHT_EDA*zE + WEIGHT_HRV*zV + WEIGHT_HR*zH  # 0.5 / 0.3 / 0.2
```

The EDA z-score zE = (phasic_eda − eda_mean) / eda_sigma. Expresses the current phasic EDA in units of its own resting spread. This is why z-scoring is essential: raw EDA is in microsiemens, HR and HRV are in percent — three different scales. Without normalising each signal to its own spread, the one with the largest raw numbers would dominate the score regardless of the 0.5/0.3/0.2 weights. After z-scoring all three speak in the same 'standard deviations from baseline' language, and the weights mean what they say.

The instantaneous score is smoothed over a short rolling buffer, then classified against thresholds derived from the baseline score distribution:

```
if s_t >= thresh_high:   state = "ultra-stressed"; unity_cmd = "decrease"
elif s_t >= thresh_mild: state = "stressed";       unity_cmd = "neutral"
else:                    state = "calm";           unity_cmd = "increase"
```

Below the lower threshold the patient is calm and the VR balloon ascends (exposure advances); between the thresholds it holds; above the upper threshold the balloon descends (exposure backs off). Thresholds sit at 1.28σ (mild, ~90th percentile of baseline) and 2.33σ (high, ~99th percentile) above the baseline score mean.

■ ***Reference.*** *The weighted-sum fusion and the 0.5/0.3/0.2 weights are a design decision of the lab/pipeline. The practice of z-scoring heterogeneous physiological channels before fusion is standard in affective-computing and multimodal arousal research. The 1.28/2.33 sigma thresholds correspond to the 90th and 99th percentiles of a normal distribution (Task Force, 1996; standard statistical convention).*

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8. Known limitations and failure modes

- **Resampler edge artifact (~0.5% of ticks).** nk.signal_resample intermittently produces a spurious ~0 as its last output sample. NeuroKit's zero-phase filter rings backward across the tail in response, and phasic[-1] can read the ringing (4-15 µS). The plausibility clamp (EDA_PHASIC_MAX_US = 1.0) catches and neutralises these events; the underlying resampler bug is documented and left unfixed because the repair heuristics tested were found to risk misfiring on real data.
- **The first ~5 s is blind.** compute_phasic_eda returns 0.0 until the buffer holds at least 5 s of data. During this period the EDA term contributes nothing to the score (HR still responds from the start; HRV from ~60 s).
- **Baseline-quality dependence for sigma.** eda_sigma captures resting phasic variability from the baseline. If the baseline ECG/EDA recording is poor or the subject moves continuously, sigma may be inflated or deflated and the floor or ceiling values deserve rechecking against more recordings.
- **Band placement is not yet validated.** The pipeline correctly separates aroused from calm recordings, but whether a given composite score level corresponds to the right absolute arousal level needs a recording with a known graded arousal protocol.
- **EDA is movement-sensitive.** Unlike ECG, EDA has no artifact-correction equivalent to Kubios. Cable touches, hand movements, or electrode slippage appear as large EDA transients. The 3-sigma cleaning on the tonic mean and the phasic plausibility clamp provide partial protection, but cannot cover all movement artifacts.

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9. References

[1] Boucsein, W. (2012). *Electrodermal Activity* (2nd ed.). Springer. — Tonic/phasic EDA physiology, SCR amplitude ranges, measurement conventions.

[2] Dawson, M. E., Schell, A. M., & Filion, D. L. (2017). The electrodermal system. In J. T. Cacioppo, L. G. Tassinary, & G. G. Berntson (Eds.), *Handbook of Psychophysiology* (4th ed.). Cambridge University Press. — Sympathetic innervation and EDA as a sympathetic arousal index.

[3] Makowski, D., Pham, T., Lau, Z. J., et al. (2021). NeuroKit2: A Python toolbox for neurophysiological signal processing. *Behavior Research Methods* 53:1689-1696. — Library providing signal_resample, eda_clean, and eda_phasic.

[4] Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology (1996). Heart rate variability: standards of measurement, physiological interpretation, and clinical use. *Circulation* 93(5):1043-1065 / *Eur Heart J* 17(3):354-381. — Standard sigma threshold conventions used in the score classification.